

THEORETICAL QUESTIONS

Question 1: What is Data Loading in Power BI?

Data Loading is the process of importing data from different sources (Excel, CSV, databases, etc.) into Power BI.

Why it is the first step:

- No analysis can happen without data
- It brings raw data into Power BI for processing
- It enables further steps like cleaning, modeling, and visualization

Question 2: Difference between “Load” and “Transform Data”

Feature	Load	Transform Data
Purpose	Directly import data	Clean & modify data before loading
Speed	Faster	Takes time
Editing	No changes	Full data cleaning options
Tool Used	Power BI	Power Query Editor

Simple idea:

- **Load** = bring data as it is
- **Transform Data** = clean data before using

Question 3: Fact Table vs Dimension Table

Fact Table

- Contains **numerical/measurable data**
- Used for calculations

Example (from COVID dataset):

- Confirmed cases
- Deaths
- Recovered

Dimension Table

- Contains **descriptive data**
- Used for filtering/grouping

Example:

- Country
- State
- Date

Example:

- Fact: Confirmed Cases = 5000
- Dimension: Country = India

Question 4: Why Star Schema is preferred?

Star Schema = One central fact table + multiple dimension tables

✓ Advantages:

- Faster performance
- Easy to understand
- Better for Power BI visuals
- Simplifies relationships

Compared to Snowflake:

- Snowflake is more complex (normalized tables)
- Star schema is simpler and faster

Question 5: Remove duplicate records (Date, Country, State)

Steps in Power Query:

1. Click **Transform Data**
2. Select columns:
 - Date
 - Country
 - State
3. Go to **Home** → **Remove Rows** → **Remove Duplicates**

	A ^B _C Country	A ^B _C State	1 ² ₃ Date
1	Brazil	State_A	43831
2	UK	State_B	43832
3	Italy	State_B	43833
4	USA	State_B	43836
5	Spain	State_C	43843
6	India	State_A	43852

Question 6: Replace null values in Vaccination_status

Steps:

Select column Vaccination_status

Go to Transform → Replace Values

Replace:

null → "Not Vaccinated" (or any default)

Hospitalized

A^B_C Vaccination_Status

10288	No
17709	No
17543	Yes
4650	Not Vaccinated
4987	Not Vaccinated
15200	Yes
4697	No
16724	Not Vaccinated
3828	No
18352	Yes
17087	No
12605	Not Vaccinated
5482	Not Vaccinated
18184	No
5463	Yes
13860	No
4912	Yes
4648	No
855	Not Vaccinated
2866	Yes
11834	Yes
16294	Yes
16566	Not Vaccinated
3112	No
4492	Not Vaccinated
15114	Yes
1517	No

PROPERTIES

Name

coronavirus_2020_dataset

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

✕ Replaced Value

Question 7 : Create a new column to calculate Recovery Rate.

Custom Column

Add a column that is computed from the other columns.

New column name

Custom

Custom column formula ⓘ

= [Recovered_Cases] / [Confirmed_Cases] * 100

Available columns

Date

Confirmed_Cases

Recovered_Cases

Deaths

Active_Cases

Tests_Conducted

Hospitalized

Vaccination_Status

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

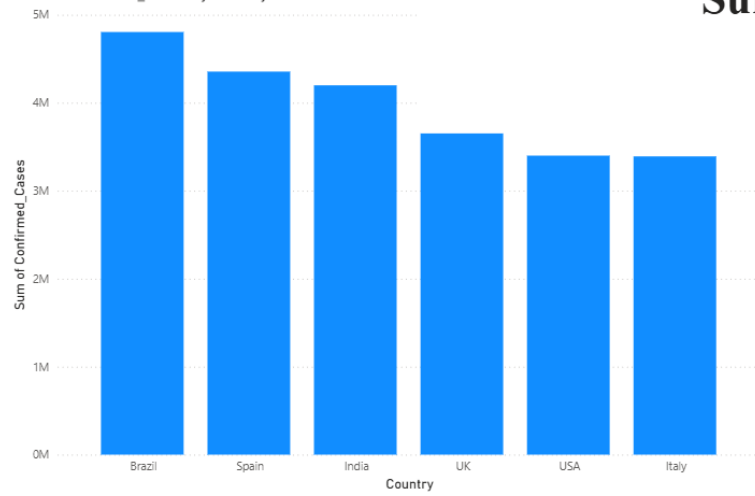
OK

Cancel

Hospitalized	Vaccination_Status	recovery rate
10288	No	136.1129971
17709	No	73.10606513
17543	Yes	114.3835616
4650	Not Vaccinated	82.30164649
4987	Not Vaccinated	41.81539559
15200	Yes	92.77808715
4697	No	353.1200283
16724	Not Vaccinated	92.59413877
3828	No	137.3030048
18352	Yes	1.863425394
17087	No	55.71795897
12605	Not Vaccinated	111.2932257
5482	Not Vaccinated	72.52504228
18184	No	77.46933831
5463	Yes	128.8736824
13860	No	173.9043
4912	Yes	22.70330724
4648	No	50.229006
855	Not Vaccinated	92.61571166
2866	Yes	80.75110216
11834	Yes	72.17482706
16294	Yes	49.37351846
16566	Not Vaccinated	153.2395695
3112	No	280.0629291
4492	Not Vaccinated	161.2341632
15114	Yes	738.721374
1517	No	141.4127392

Question 8 : Create a summarized table showing total confirmed cases by Country

Sum of Confirmed_Cases by Country



Summary

Sum of Confirmed_Cases	Country
4804531	Brazil
4198680	India
3390760	Italy
4354613	Spain
3651598	UK
3399318	USA
23799500	